

Random Variables: Functions and Independence

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MATH 241

Week 6, Lecture #2 — Ch. 3.7, 3.8, 3.11

Checkpoint!!

You will be working in groups of 2-3 for this checkpoint.

- 15 minutes to work on it!
- Free to use your notes, lecture material, and textbook as a reference.
- Discuss the questions, answer them, and then turn in the worksheet when you're finished.

Reminders!

- Homework #4 due on Sunday at 11:59pm
- Office hours on Friday from 10-11:30am

Today's Material

- Walk-through example from Tuesday
- Functions of RVs
 - ▶ How do we use them? What are its uses?
- Independence of RVs
 - ▶ Benefits?
- R!!!
 - ▶ How can I use R to solve probabilities?

Examples

- ② Brandon decides to enter the lottery repeatedly for hiking Angels Landing in Zion National Park, hoping he'll eventually be able to hike the popular trail. The chance of being approved to hike Angels Landing is 25%. Find the probability that:
- ▶ in 10 independent drawings, he will be able to have the opportunity to hike at least twice.
 - ▶ he will have a chance to hike within 3 drawings.

Definition (3.7.1)

For an experiment with sample space $\mathcal{S}$, an RV X , and a function $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(X)$ is the RV that maps s to $g(X(s))$ for all $s \in \mathcal{S}$.

In other words: a composition of functions X and g , meaning:

- 1 Map outcomes of $\mathcal{S}$ to X
- 2 Map outcomes of X to g

Example: Winnings with a Dice Roll

Suppose I roll a standard die and I get 2 times the face value minus 5 dollars for the face value I get. Let X be the face value of the roll, and Y be my winnings for that value.

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- Note that $Y = 2X - 5$
- To find the PMF of Y , I can use the PMF of X :

$$P(Y = y) = P(2X - 5 = y) = P\left(X = \frac{y + 5}{2}\right) = \frac{1}{6}$$

Note the values of my support: $y \in \{-3, -1, 1, 3, 5, 7\}$

Theorem 3.7.3

Theorem

Let X be a discrete random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$. Then, the support of $g(X)$ is the set of all y such that $g(x) = y$ for at least one x in the support of X , and the PMF of $g(X)$ is:

$$P(g(X) = y) = \sum_{x: g(x)=y} P(X = x)$$

Functions of Two RVs

We can also have functions of two random variables:

Definition

Given an experiment with sample space $\mathcal{S}$, if X and Y are RVs that map $s \in \mathcal{S}$ to $X(s)$ and $Y(s)$ respectively, then $g(X, Y)$ is the RV that maps s to $g(X(s), Y(s))$.

Example: Dice Rolls

Let X be the number on the first die and Y be the number on the second die. I might be curious about the following:

- $X + Y$: the sum of the two dice
- $\max\{X, Y\}$: the highest face value of the two dice
- $Y - X$: the difference between the second and first die

Can use tabulation to get values of $g(X, Y)$ and then find my PMF.

Independence of RVs

We've worked with independence of events – the concepts also translate to random variables:

Definition (3.8.1)

Random variables X and Y are said to be **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y),$$

for all $x, y \in \mathbb{R}$.

- In discrete case: $P(X = x, Y = y) = P(X = x)P(Y = y)$

Independence of Many RVs

Definition (3.8.2)

Random variables $X_1, \dots, X_n$ are said to be **independent** if:

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = P(X_1 \leq x_1) \times \dots \times P(X_n \leq x_n),$$

for all $x_1, \dots, x_n \in \mathbb{R}$.

- In discrete case:

$$P(X_1 = x_1, \dots, X_n = x_n) = P(X_1 = x_1) \times \dots \times P(X_n = x_n)$$

- In many cases, we might care about many random variables that are independent AND identically distributed...

Definition 3.8.6: Independent and Identically Distributed (IID)

Random variables that are independent and have the same distribution are called...

- **Independent** and **identically distributed**
- i.e., IID

Examples:

- **IID:** Two students take same exam of 5 multiple-choice questions and pick answers randomly. X and Y are the number of correct answers students A and B get.
- **Independent, but not identical:** X is the number of correct answers student A gets, Y is the roll of a die.
- **Neither independent nor identical:** X is the number of correct answers student A gets and Z is the number of incorrect answers student A gets

Example of IID

Suppose we have 10 coin tosses and care about whether we get a heads on each of the coin tosses. Each outcome can be described as X_i , which is distributed as a Bernoulli distribution, with $p = 0.5$.

- What is the distribution of the sum of heads in 10 coin tosses?

Example of IID

Suppose we have 10 coin tosses and care about whether we get a heads on each of the coin tosses. Each outcome can be described as X_i , which is distributed as a Bernoulli distribution, with $p = 0.5$.

- What is the distribution of the sum of heads in 10 coin tosses?
 - ▶ $Y = X_1 + \cdots + X_n \sim \text{Bin}(n = 10, p = 0.5)$
- The X_i s are independent and identically distributed
 - ▶ Because of this, we can establish a property: Y can be rewritten as the sum of X_i s, where $X_i \sim \text{Bern}(p)$

Connections between Random Variables

You will find that a function of certain random variables that are commonly distributed in one way, given some properties (e.g. independence, identical distribution), will result in a random variable that is commonly distributed in another way.

- Bernoulli to binomial?
 - ▶ $Y = X_1 + \cdots + X_n$
- Helps for future calculations, especially with given parameters

Next week: expectations!

- Focus on the average or mean
 - ▶ $E(X)$
 - ▶ Solve for our own discrete and common distributions
- How does it translate to the variance?

Let's work through some R code!